

Background

Lupines (*Lupinus ssp.*) is a legume family of the Fabaceae, is an example of a plant that offers numerous benefits for both agriculture and food production but has not been widely used¹ or extensively studied². Compared to other legumes, lupin comprises a wide variety of species (about 300), while only four of them are considered cultivated species and have been studied due to their nutritional properties³. Therefore, the study of wild species and the examination of their nutritional potential is highly significant in order to promote the expansion of unutilized plant-based food sources. This research focuses on characterizing of the two common wild species in Israel: *Lupinus pilosus* and *Lupinus palaestinus* in terms of nutritional and functional potential, examining the whole seed and the proteins that compose it.

Methodology

Lupin Preparation

Lupin seeds from Volcani Institute plots were freeze-dried and ground into powder.

Chemical and nutritional analysis

Carbohydrate and protein contents, mineral profile, total phenolics (TPC), and antioxidant activity (via FRAP and DPPH) were analyzed.

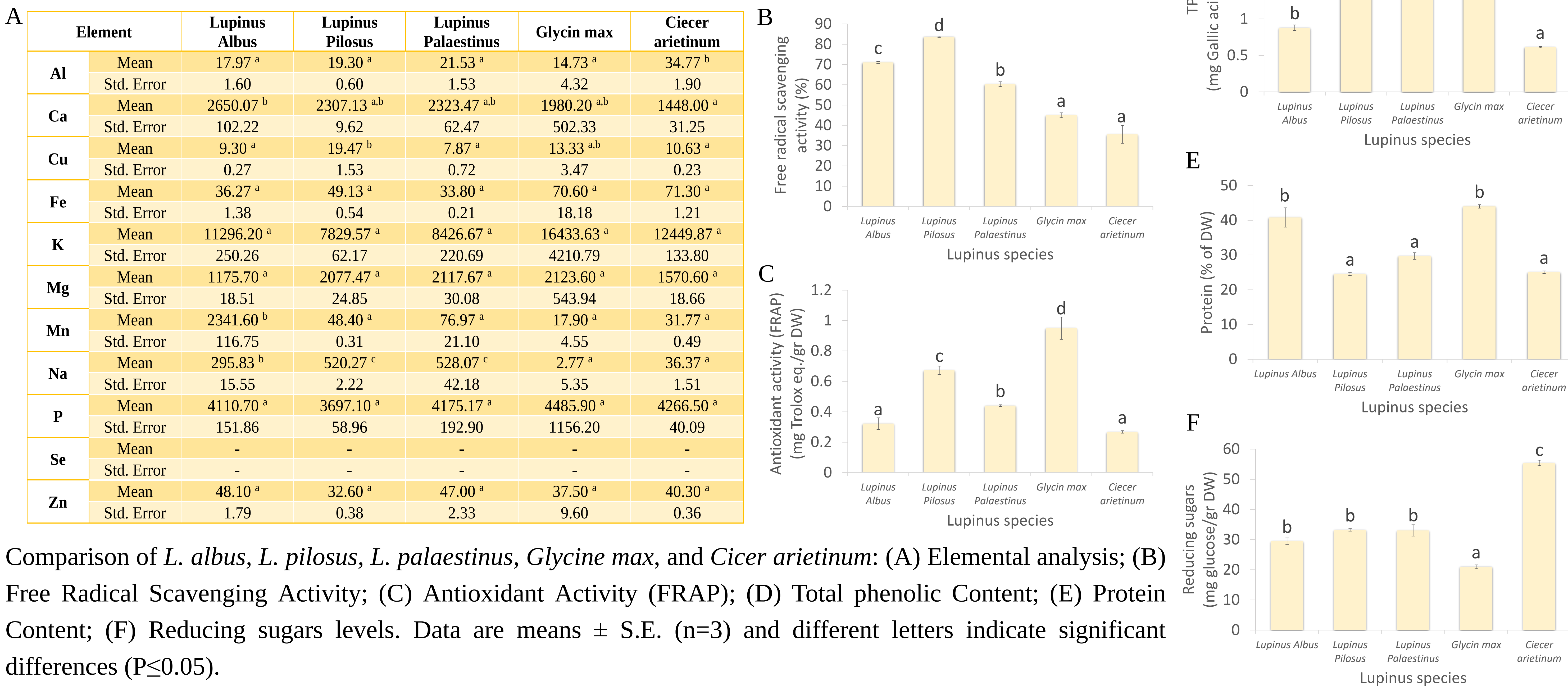
Protein fractions Extraction

Proteins were extracted by pH-based precipitation to separate fractions by isoelectric point.

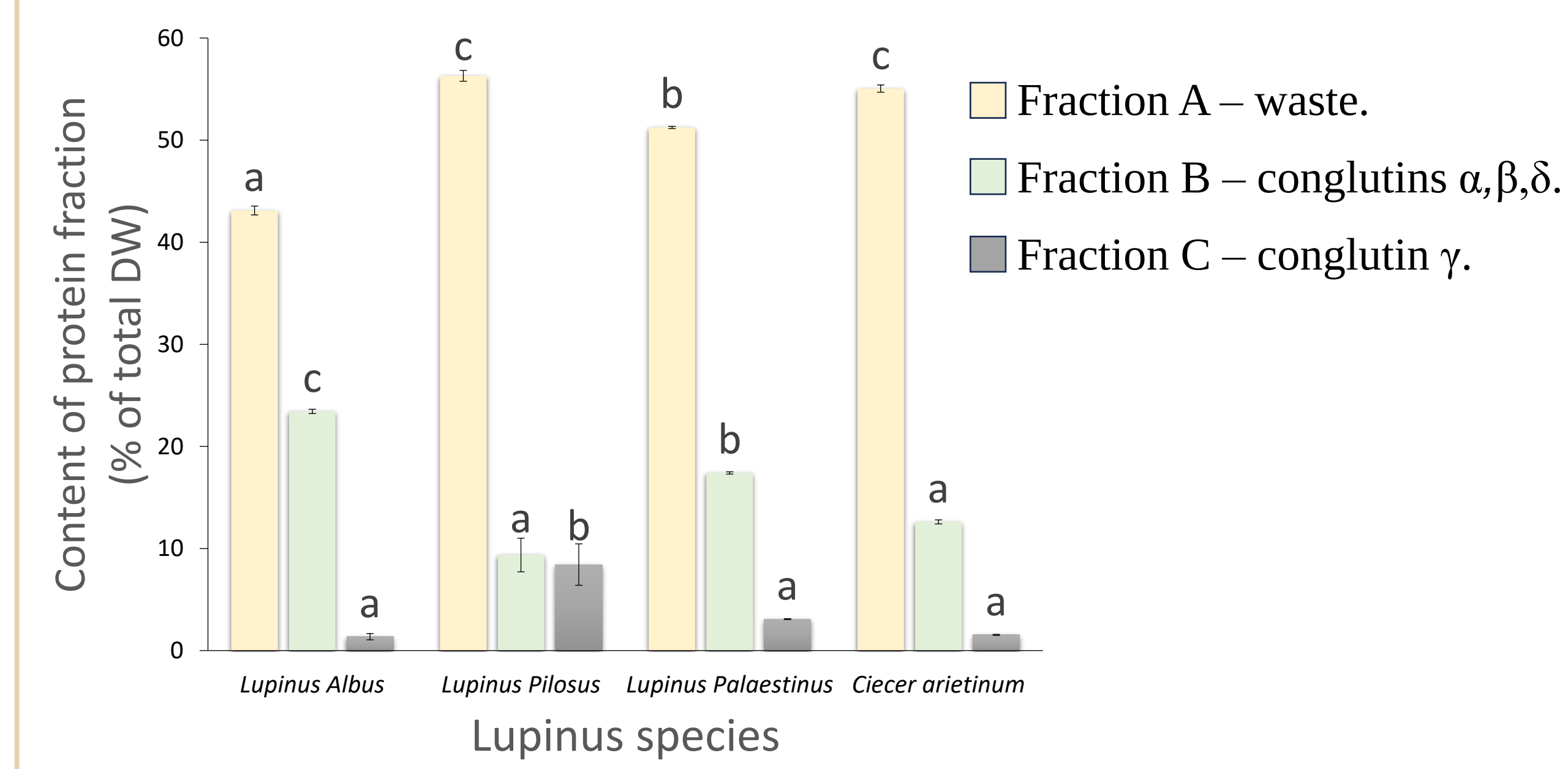
Results

Chemical and nutritional analysis

Results of nutritional characterization of the wild lupin species *L. pilosus* and *L. palaestinus* compared to the cultivated lupin species *L. albus*, with *Cicer arietinum* (chickpea) and *Glycin max* (soybean) examined as references.

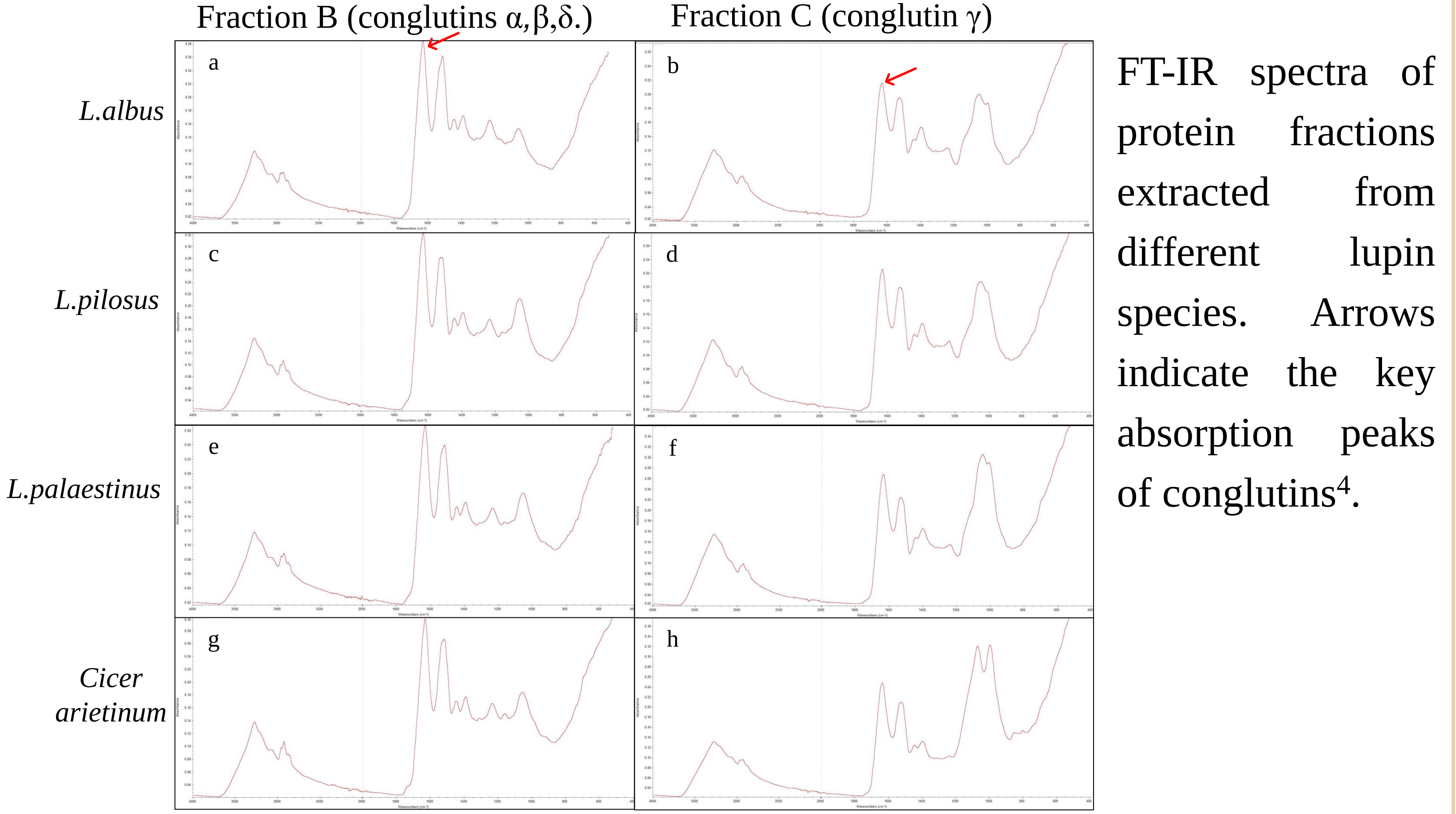


Protein fraction Extraction - Quantitative results



The fraction percentage from whole lupin seed on a dry basis, obtained through the protein extraction process. Data are means ± S.E.(n=3) and different letters indicate significant differences (P≤0.05).

Protein fraction Extraction - Qualitative results



Conclusion

Wild lupin species demonstrate comparable, and in some aspects superior, nutritional quality to cultivated lupin and chickpea. However, their protein content does not surpass that of the cultivated variety, which might be attributed to breeding processes. Cultivating *L. palaestinus* and *L. pilosus* could enhance plant-based food sources and provide valuable ingredients. The quantitative results of the protein fractions reveals significant variations in conglutin profiles between wild and cultivated lupins, while the qualitative analysis confirms the similarity of the tested lupins to the literature IR fingerprint of conglutins. Further steps in the reasearch aim to characterize the functionality of whole lupin flour and their protein fractions for potential applications in food.

Bibliography

- Shrestha, S., Hag, L. van 't, Haritos, V. S. & Dhital, S. Lupin proteins: Structure, isolation and application. Trends Food Sci Technol 116, 928–939 (2021).
- Bryant, L., Rangan, A. & Grafenauer, S. Lupins and Health Outcomes: A Systematic Literature Review. Nutrients 14, 327 (2022).
- Khan, M. K., Karnpanit, W., Nasar-Abbas, S. M., -E-Huma, Z. & Jayasena, V. Review Phytochemical composition and bioactivities of lupin: a review. (2015) doi:10.1111/ijfs.12796.
- Tang, J., Cases, L., Alves, S., Sun, D.-W. & Tiwari, B. K. Protein extraction from lupin (*Lupinus angustifolius* L.) using combined ultrasound and microwave techniques: Impact on protein recovery, structure, and functional properties. Ultrason Sonochem 115, 107232 (2025).